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MILITARY SPECIFICATION

LUBRICATING OIL, INSTRUMENT, -65° TO +400° F

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one grade of aircraft instrument oil for use in the temperature range of -65° to +400° F.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

P-D-680	Dry Cleaning Solvent
QQ-S-624 —	Steel Bar, Alloy, Hot Rolled and Cold Finished (General Purpose)
QQ-S-698	Steel, Sheet and Strip, Low-Carbon
QQ-S-766	Steel Plates, Sheets, and Strip Corrosion Resisting

Military

MIL-S-13282	Silver and Silver Alloys

STANDARDS

Federal

Fed. Test Method Std. No. 791	Lubricants, Liquid Fuels, and Related Products; Methods of Testing

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-290	Packaging, Packing, and Marking of Petroleum and Related Products
MIL-STD-831	Test Reports, Preparation of

FSC 9150

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM Standards on Petroleum Products and Lubricants

(Copies of ASTM publications may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Qualification.- The lubricating oil furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable Qualified Products List.

3.2 Materials.- The lubricating instrument oil shall consist substantially of synthetic base oil and additive materials to impart oxidation stability and corrosion-protective properties. The composition of the lubricant is not limited, but a formulation composed substantially of nonpetroleum materials will be required to meet this specification.

3.3 Viscosity.- The viscosity of the oil shall not exceed 5,000 centistokes at -65° F and shall be not less than 2.5 centistokes at +400° F, when tested as specified in 4.5.1.

3.4 Low-temperature stability.- There shall be no gelling or separation of solid or liquid phases in the oil after storage at a temperature not higher than -65° F for 72 hours, when tested as specified in 4.5.1. The presence of a dense cloud which does not settle out shall not be cause for rejection.

3.5 Pour point.- The pour point of the oil shall be not higher than -80° F, when tested as specified in 4.5.1.

3.6 Flash point.- The flash point of the oil shall be not less than 500° F, when tested as specified in 4.5.1.

3.7 Fire point.- The fire point of the oil shall be not less than 550° F, when tested as specified in 4.5.1.

3.8 Acid number.- The acid number of the oil shall be not greater than 0.1, when tested as specified in 4.5.1.

3.9 Corrosion and oxidation stability.- The oil shall comply with the following requirements, when tested as specified in 4.5.1.

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3.9.1 Corrosion.- Steel test specimens conforming to QQ-S-624, QQ-S-698, and QQ-S-766, and a silver test specimen conforming to MIL-S-13282 shall be arranged to form a tetrahedron with sides fastened together, using fiberglass thread. The change in weight of each steel and silver test specimen shall be not greater than ±0.2 milligrams per square centimeter when subjected to the oils for 72 hours at 400° F. There shall be no pitting, etching, or visible corrosion on the surface of any of the metals when viewed under 20X magnification. Slight staining of the metals will be permitted but dark gray or black deposits on the metals shall be cause for rejection.

3.9.2 Oxidation.- The oil shall not have changed more than ±5 percent from the original viscosity in centistokes at 100° F after the oxidation-corrosion test. The neutralization number change shall be not greater than 0.5 after oxidation. There shall be no evidence of separation or gumming of the oil.

3.10 Evaporation.- There shall not be more than 1.5 percent, by weight, evaporation loss after the oil has been subjected to the evaporation test for 22 hours at 400° F, when tested as specified in 4.5.1.

3.11 Corrosivity.- When subjected to the test specified in 4.5.1, test areas of at least two of the three disks shall show no evidence of corrosion, pitting, or other attack. The third disk may show no more than three spots within the area covered by the clip.

3.12 Steel-on-steel wear test.- The wear scar diameter shall not exceed 1.0 mm. at a 10-kilogram load, when tested as specified in 4.5.2.3 at 400° F.

3.13 Storage stability.- A sample of the test fluid after storage in the dark for 6 months shall exhibit no formation of resinous gums, sludges, or insoluble materials or separation, when tested as specified in 4.5.3.

3.14 Workmanship.- The oil shall contain no admixture of resins, rubber, gums, fatty oils, oxidized hydrocarbons, or any other additive unless specifically approved by the activity responsible for qualification. The oils shall be entirely homogeneous and free from lumps of undissolved additive, water, dirt, or sediment, and immediately before final packaging the oil shall be filtered through a blotter press.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections.- The examining and testing of the oil shall be classified as follows:

(a) Qualification inspections (4.3)
(b) Quality conformance inspections (4.4)

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4.3 Qualification inspections.- The qualification inspections shall consist of all of the examinations and tests of this specification, excluding the inspection of filled containers (4.5.4).

4.3.1 Test sample.- The test sample shall consist of 1 gallon of oil. The sample shall be identified by securely attached, durable tags marked with the manufacturer's own number and any additional information required by the letter of authorization.

4.3.2 Test report.- The qualification sample shall be accompanied by a certified test report prepared in accordance with MIL-STD-831 showing the results of all of the qualification tests required by this specification, with the exception of storage stability (4.5.3).

4.4 Quality conformance inspections.- Tests for quality conformance inspection of individual lots shall consist of tests for all requirements specified in section 3, with the exception of storage stability as specified in 3.13.

4.4.1 Sampling.-

4.4.1.1 Lot.- A lot shall consist of all the oil manufactured as one batch and presented for acceptance at one time.

4.4.1.1.1 Batch.- A batch shall be the end product of all raw materials mixed or blended in a single operation.

4.4.1.2 Samples.- Samples shall consist of 1 gallon of oil from a single lot, selected in accordance with Method 8001 of Fed. Test Method Std. No. 791.

4.4.1.3 Inspection.- Inspection shall be in accordance with Method 9601 of Fed. Test Method Std. No. 791. If any specimen fails to meet the specified test requirements, the lot or batch shall be rejected.

4.4.1.4 Sampling for inspection of filled containers.- A random sample of filled containers shall be selected from each lot in accordance with MIL-STD-105, inspection level I, and an AQL of 2.5 percent defective. The sample containers shall be subjected to the inspection of filled containers as specified in 4.5.4.

4.5 Test methods.-

4.5.1 The tests of this specification shall be conducted in accordance with the applicable test methods of Fed. Test Method Std. No. 791 and ASTM Standards on Petroleum Products and Lubricants, as listed in table I, and other methods as described in 4.5.2 and 4.5.3.

4.5.2 Steel-on-steel wear test.-

4.5.2.1 Apparatus.- The apparatus for this test shall consist of a Shell four-ball wear tester (see 6.4), an Alnor temperature controller, 1/2-inch steel balls (440C stainless steel, for use with the four-ball tester), and wrenches supplied with the four-ball tester.

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TABLE I. Test methods

Test	Fed. Std. No. 791 Method No. 1/	ASTM Method No.
Viscosity (-65° F and +400° F)	305	D445
Low-temperature stability	3458	---
Pour point	201	D97
Flash point	1103	D92
Fire point	1103	D92
Acid number	5105	D974
Corrosion and oxidation 2/	5308	---
Evaporation 3/	351	D972
Corrosivity	5322	---

1/ Where a corresponding ASTM number is listed, the ASTM method is to be used in conducting the test.
2/ The test temperature shall be 400° F, the duration of the test shall be 72 hours; the metal specimens used shall be those listed in 3.9.1.
3/ The test temperature shall be 400° F.

4.5.2.2 Preparation.- The steel balls, chuck and ball pot assembly shall be individually and thoroughly washed in dry cleaning solvent conforming to P-D-680, rinsed in clean solvent, swished in petroleum ether (ligroin) and then dried by allowing the solvent to evaporate. Four clean balls shall be selected. One ball shall be placed in the revolving chuck ball holder. The chuck shall be firmly tightened and placed in the machine. Three balls shall be placed in the ball pot assembly and locked in by tightening the cup sufficiently to prevent any movement or rotation of the balls during the test. The secured balls in the ball pot assembly shall be coated with an ample covering of the test oil and the ball pot assembly placed in position in the machine.

4.5.2.3 Test procedure.- The speed of the tester shall be adjusted to 600 revolutions per minute (rpm), the 10-kilogram load applied, and the mercury cup assembly moved up to make contact with the heater and thermocouple terminals of the ball pot assembly. The temperature controller shall be adjusted to 77° F, the heater switch turned on, and the variable transformer dial adjusted until the unit has lined out at a test temperature of 400° F. The motor shall then be turned on. At the end of 60 minutes, the motor and heater switches shall be turned off, the mercury cup lowered, and the load released. The ball pot assembly shall be removed from the test machine, flushed thoroughly with petroleum ether to remove the oil sample, and dried with a clean, soft cloth. To facilitate locating the scars when the balls are viewed under a microscope, a circle shall be inscribed with a diamond tip pencil around the scar area of each of the three balls. The balls shall then be removed from the ball pot assembly, cleaned in naphtha and petroleum ether conforming to ASTM D91, and placed in a suitable holder (a smaller wooden block) on a microscope stage. The minimum diameter of the scar and the diameter at 90 degrees to this position shall be measured on each of the three balls. The average diameter, based on six readings, shall be calculated and reported in millimeters. The torque measuring device, which is part of the equipment, shall not be used.

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4.5.3 Storage stability.- Three 4-ounce samples of the test fluid shall be placed in separate airtight glass containers and stored in the dark at temperatures between 65° and 90° F for 6 months. No agitation of the samples will be permitted during storage of the sample. The samples shall then be examined visually for conformance to 3.13.

4.5.4 Inspection of filled containers.- Each sample filled container selected in accordance with 4.4.1.4 shall be examined to verify compliance with all packaging, packing, and marking requirements specified herein.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- The oil shall be packaged, packed, and marked in accordance with MIL-STD-290. The type and size of containers and level of packaging and packing shall be as specified by the procuring activity (see 6.2). The marking nomenclature shall be:

LUBRICATING OIL, INSTRUMENT, -65° TO +400° F

6. NOTES

6.1 Intended use.- The oil is intended for use in aircraft instruments such as tachometer generators, gyro motors, gyro gimbals, and other applications involving light to moderate loads, small oscillatory motions, and relatively high speeds (up to 24,000 rpm).

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Quantity desired.
(c) Applicable levels of packaging and packing (see 5.1).
(d) Type and size of containers (see 5.1).

6.3 Qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Air Force Materials Laboratory, Attn: MAAE, Wright-Patterson Air Force Base, Ohio 45433, and information pertaining to qualification of products may be obtained from that activity.

6.4 Source of apparatus.- Shell four-ball wear testers may be obtained from the Precision Scientific Company, Chicago, Illinois, Catalog No. 73603.

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Custodians:
Army - MR
Navy - WP
Air Force - (11)

Reviewer activities:
Army - MI, MR, MU
Navy - WP
Air Force - (11), (68)

User activities:
Army - MO, SM
Navy - MC

Preparing activity:
Air Force - (11)

Project No. 9150-0102

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS.

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

MIL-L-27694A Lubricating Oil, Instrument, -65° to +400° F

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ Direct Government Contract ☐ Subcontract

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)

SUBMITTED BY (Printed or typed name and activity) DATE

DD Form 1426

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